

Markdown Parser in PostScript

A *complete* Markdown processor written entirely in **PostScript** â a stack-based page-description language not traditionally associated with text parsing. The parser handles both block-level structure and inline character-level markup with no external tools or preprocessing.

Architecture

Block Dispatcher

Each line in `MDLines` is passed to `processLine`, which classifies it by its leading characters. The classifier is a nested `ifelse` chain operating in priority order so that `###` is matched before `##`, and `##` before `#`.

Inline Scanner

The `renderInline` procedure implements a character-level state machine. It walks the string one byte at a time, accumulating non-space bytes into a 512-byte word buffer. On hitting a space it flushes the buffer via `showWord`; on hitting `it` toggles the appropriate style flag and calls `selectFont` to switch the active PostScript font.

Word-Wrap Engine

`showWord` measures each word with `stringwidth` before placing it. If `CurX + spaceW + wordW` would exceed the right margin, a new line is started via `bodyNL`, which resets `CurX` and decrements `CurY`. When `CurY` drops below the bottom margin, `advanceY` calls `showpage` and initialises a fresh page.

Supported Elements

Inline Spans

These inline styles may be freely nested and combined within any block:

- **Bold** via `double asterisks`
- *Italic* via `single asterisks`
- `Inline code` via backtick pairs â rendered in Courier on a grey pill
- **Bold and italic** can be nested inside each other

Block Elements

1. Headings at three levels (H1, H2, H3) with typographic rules
2. Paragraphs with automatic word-wrap and inter-paragraph spacing
3. Unordered lists with filled-circle bullets in the indent margin
4. Ordered lists with bold numeric labels flush-right to the indent
5. Blockquotes with a coloured left bar and tinted background
6. Fenced code blocks preserving verbatim layout in Courier
7. Horizontal rules drawn as hairlines across the full column

Horizontal Rules

A horizontal rule is produced by three or more `-`, `,` or `_` chars.

Simplicity is prerequisite for reliability. The elegance of a PostScript solution lies in the stack discipline: every procedure leaves the stack exactly as clean as it found it.

Fenced Code Block

```
% PostScript Markdown inline scanner (core loop)
ri s ri i det /ri c exch def
ri c 42 eq {                                % asterisk
  ri s ri i 1 add det 42 eq {
    flushWord /TnBold TnBold not def
    /ri i ri i 2 add def
  } { flushWord /TnItalic TnItalic not def
    /ri i ri i 1 add def } ifelse
} { ri c 96 eq {                             % backtick
  flushWord /TnCode TnCode not def
  /ri i ri i 1 add def
} { ri c 32 eq {                             % space
  flushWord /ri i ri i 1 add def
} { ri c appendChar                         % accumulate
  /ri i ri i 1 add def
} ifelse } ifelse } ifelse
```

Design Notes

Font Metrics

All nine fonts are pre-scaled with `scalefont` at startup and stored in named variables (`FBody` , `FBold` , `FItal` , `FBoldI` , `FMono` , `FMonoB` , `FH1` , `FH2` , `FH3`). This avoids repeated font lookups inside the inner rendering loop, where `selectFont` is called for every style transition.

State Reset

At the end of `renderInline` all three toggle flags are reset to `false` regardless of whether the source Markdown had balanced markers. This prevents style leakage across lines, since each logical Markdown line is processed as an independent call to `renderInline` .

Page Breaks

`advanceY` checks `CurY` against a bottom threshold that reserves space for the footer rule. When the threshold is crossed mid-paragraph, `newPage` emits `showpage` , redraws the chrome (background, footer), and resets the cursor to the top of the fresh page. The next word lands at the top-left of the new column.

References

- Gruber, J. (2004). *Daring Fireball: Markdown*. daringfireball.net
- CommonMark Spec (2023). *CommonMark 0.31*. spec.commonmark.org
- Adobe Systems (1999). *PostScript Language Reference* , 3rd edition.
- Knuth, D.E. (1984). *The TeXbook*. Addison-Wesley.